

DP024

Physics 2

Semester 2

Session 2023/2024

1 hour 45 minutes



**KOLEJ
MATRIKULASI
LABUAN**

PRA PSPM

DP024 (PHYSICS2)

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

INSTRUCTIONS TO CANDIDATE:

The question paper consists of 6 questions.

Answer **all** questions.

The use of electronic calculator is permitted.

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_o	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ϵ_o	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Atomic mass unit <i>Unit jisim atom</i>	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law <i>Pemalar hukum Coulomb</i>	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure <i>Tekanan atmosfera</i>	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water <i>Ketumpatan air</i>	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

1. $a = -\omega^2 y$	19. $C = \frac{Q}{V}$
2. $y = A \sin \omega t$	20. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$
3. $\omega = \frac{2\pi}{T} = 2\pi f$	21. $C_{eff} = C_1 + C_2 + C_3 + \dots + C_n$
4. $v = A\omega \cos \omega t$	22. $\tau = RC$
5. $a = -A\omega^2 \sin \omega t$	23. $Q = Q_o e^{\frac{-t}{RC}}$
6. $k = \frac{2\pi}{\lambda}$	24. $Q = Q_o \left(1 - e^{\frac{-t}{RC}} \right)$
7. $y(x, t) = A \sin(\omega t \pm kx)$	25. $B = \frac{\mu_o I}{2\pi r}$
8. $v = \frac{dy}{dt}$	26. $B = \frac{\mu_o I}{2r}$
9. $v = f \lambda$	27. $B = \mu_o n I$
10. $y = 2A \cos kx \sin \omega t$	28. $\phi = BA \cos \theta$
11. $v = \sqrt{\frac{T}{\mu}}$	29. $\Phi = N\phi$
12. $\mu = \frac{m}{L}$	30. $\varepsilon = -\frac{d\phi}{dt}$
13. $I = \frac{dQ}{dt}$	31. $\varepsilon = Blv \sin \theta$
14. $Q = ne$	32. $\varepsilon = -NA \frac{dB}{dt}$
15. $\rho = \frac{RA}{l}$	33. $\varepsilon = -NB \frac{dA}{dt}$
16. $R_{eff} = R_1 + R_2 + R_3 + \dots + R_n$	
17. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$	
18. $V = IR$	

$$34. \quad R = 2f$$

$$35. \quad \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$36. \quad m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$37. \quad y_m = \frac{m\lambda D}{d}$$

$$38. \quad y_m = \frac{(m + \frac{1}{2})\lambda D}{d}$$

$$39. \quad \Delta y = \frac{\lambda D}{d}$$

Answer **all** questions.

- 1 (a) The equation for a mechanical wave is given as

$$y = 0.40 \sin(25t - 8x)$$

where x and y are in meters and t is in seconds. Determine the

- (i) direction of wave propagation.
- (ii) frequency.
- (iii) wavelength.
- (iv) wave speed.
- (v) particle vibrational velocity at $t = 0.1$ s and $x = 5$ cm.

[10 marks]

- (b) A sinusoidal wave travelling in a medium is represented by the equation

$$y_1 = 0.03 \sin(40\pi t - 10\pi x)$$

where x and y are in meters and t is in seconds.

- (i) Calculate the wavelength of the wave.
- (ii) Write down the equation of a second wave which would produce a stationary wave when combine with the wave y_1 .
In which direction is this second wave travelling?
- (iii) Calculate the amplitude of the stationary wave.

[6 marks]

- 2 (a) A wire of cross – sectional 0.9 mm^2 is connected to a voltage, V and the current, I in the wire is measured. The resistance of the wire is 0.45Ω and the resistivity is $1.68 \times 10^{-8} \Omega \text{ m}$. If the current measured is 0.5 A , calculate the
- number of electrons in a minute.
 - length of the wire.
 - voltage in the wire.

[6 marks]

(b)

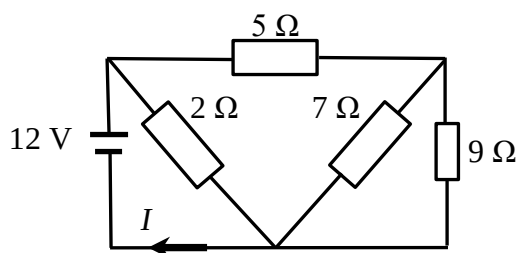
**FIGURE 2.1**

FIGURE 2.1 shows a combination of four resistors connected to 12 V battery.

Calculate the

- effective resistance.
- current in the circuit.

[6 marks]

3 (a)

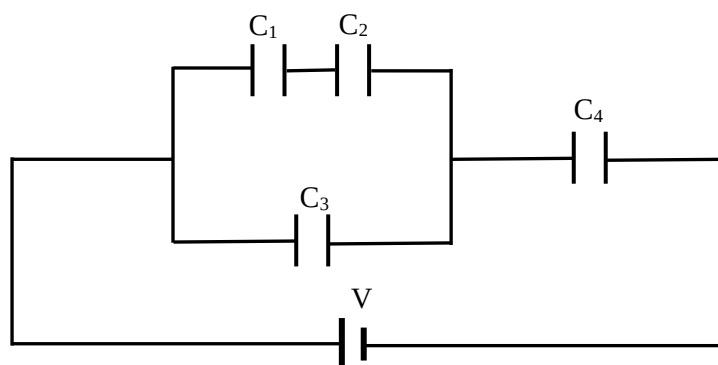


FIGURE 3.1

FIGURE 3.1 shows an arrangement of four capacitors connected to a 12 V battery.

Given $C_1 = 3 \mu\text{F}$, $C_2 = 6 \mu\text{F}$, $C_3 = 4 \mu\text{F}$, and $C_4 = 6 \mu\text{F}$. Calculate the

- (i) effective capacitance of the circuit.
- (ii) charge stored in C_4 .

[6 marks]

(b)

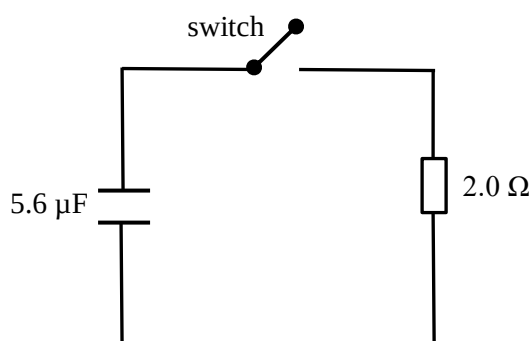


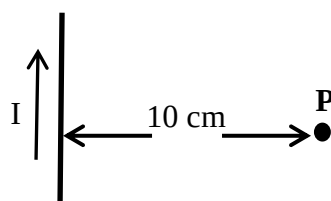
FIGURE 3.2

A fully charge $5.6 \mu\text{F}$ capacitor is connected to a 2.0Ω resistor and switch as in FIGURE 3.2. The potential difference across the capacitor is 12.0 V. Calculate the

- (i) time constant of the circuit.
- (ii) time for the charge in the capacitor to reduce one fourth of its initial value.

[4 marks]

4 (a)

**FIGURE 4.1**

An electric wire in the wall of a building carries a direct current of 25 A vertically upward as shown in **FIGURE 4.1**. What is the magnitude and direction of the magnetic field due to this current at point **P**.

[3 marks]

(b)

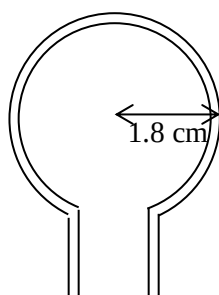
**FIGURE 4.2**

FIGURE 4.2 shows a coil of 3 turns with a resistance of $4\ \Omega$ connected to a 12 V battery. The radius of the loop is 1.8 cm.

- (i) Calculate the amount of current flowing through loop.
- (ii) Determine the magnetic field strength due to each turn at the center of the coil.

[4 marks]

- (c) A flat circular coil of 1200 turns and of mean radius 8.0 mm is connected to an ammeter of negligible resistance. The coil has a resistance of $6.3\ \Omega$. The plane of the coil is placed perpendicular to a magnetic field of flux density 0.15 T from a solenoid. The current in the solenoid is switched off. It takes 20 ms for the magnetic field to decrease from its maximum value to zero. Calculate the

- (i) average magnitude of the induced e.m.f across the ends of the coil.
- (ii) average current measured by the ammeter.

[7 marks]

- 5 (a) An object is placed at a distance of 10 cm from a spherical mirror. Image is formed 50 cm in front of the mirror.

- (i) Calculate the focal length and state the type of mirror.
- (ii) Determine the linear magnification.
- (iii) Sketch a labelled ray diagram to show the formation of the image.

[7 marks]

- (b) An upright image is formed 60 cm from the lens. The image's height is 3 times of object's height.

- (i) Calculate the object distance.
- (ii) Calculate the focal length of the lens.
- (iii) State **THREE** characteristics of image formed.

[7 marks]

- 6 (a) In a double slit experiment, the distance between the two slits is 0.45 mm. The third bright fringe is 6.85 mm from the central bright. The distance from a double-slits to the screen is 1.8 m.
- (i) Calculate the wavelength of the light.
 - (ii) Calculate the distance of the fourth dark fringe from the central bright.
 - (iii) Determine the distance between two consecutive bright fringes.
 - (iv) State the effect of the fringe separation if the separation distance between double slits is increased. Explain your answer.

[10 marks]

- (b) A mixture of red light (wavelength = 665 nm) and yellow green light (wavelength = 565 nm) incident on the double-slits separation 2.2 mm. A flat screen is located 2.15 m away from double slits. Calculate the
- (i) distance of 2nd order red fringe from the central bright.
 - (ii) distance of 5th order yellow green fringe from the central bright.
 - (iii) separation distance between 2nd order red fringe and 5th order yellow green fringe.

[4 marks]

END OF QUESTION PAPER